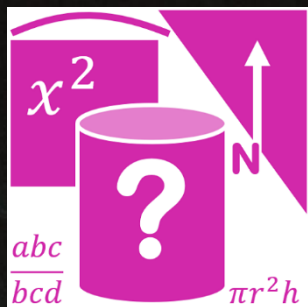




# 1.4 Mathematical Reasoning

## 91947 Greenbook



**snedmaths**



## INTRODUCTION

This workbook contains practice external examinations for (1.4) AS91267 Demonstrate mathematical reasoning. Assessment schedules are online at <https://sites.google.com/view/snedmaths/new-level-1>

Schools and teachers can use this resource however they wish, with the provisos that:

- it is unmodified and retains the link to the site in the header;
- no part of this resource is re-used as part of a site that requires a subscription or fee; and
- no part is used or modified to be a school “derived grade” or “end of topic” examination.

If you find errors, please let me know!

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Lots of my resources are online at <https://sites.google.com/view/snedmaths>

## MATHEMATICAL REASONING 91947-EXM-1

## SNEDMATHS

### QUESTION ONE

- (a) Show that the ratio of the volume of a sphere of radius  $r$  to its surface area,

$$\frac{\text{volume}}{\text{surface area}}$$

can be simplified to  $c \times r$ , where  $c$  is a constant.

State clearly the value of  $c$ .

- (b) Frankie and Sam each think of a number.

Frankie's number plus twice Sam's number is 77.

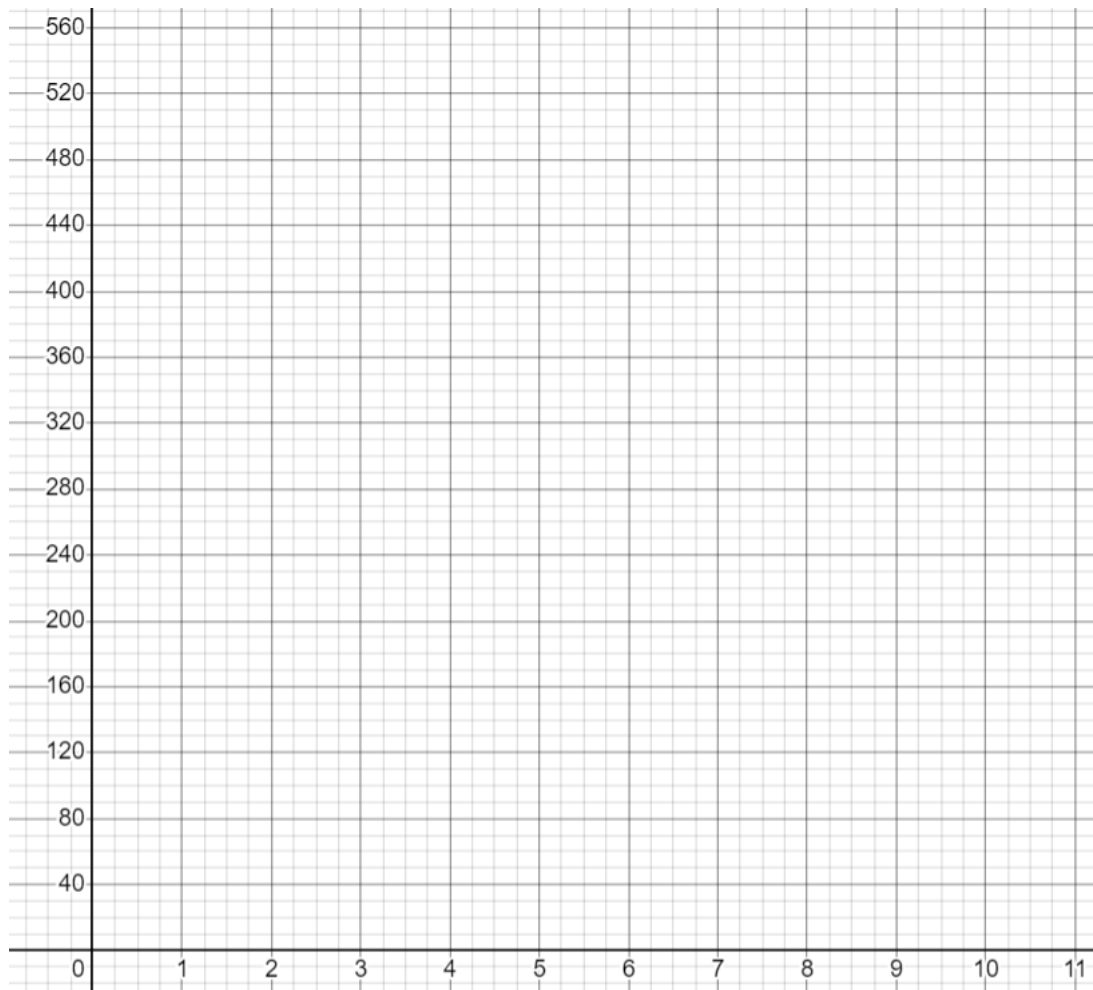
Sam's number plus twice Frankie's number is 100.

Use algebra to find their numbers.

- (c) Victor keeps partial records of his distance travelled, time travelled, and average speed on three parts of a trip.

| section of trip            | distance (km) | time               | average speed (km/h) |
|----------------------------|---------------|--------------------|----------------------|
| Tauranga to Rotorua        |               | 1 hour 15 minutes  | 52 km/h              |
| Rotorua to Napier          | 240 km        |                    | 40 km/h              |
| Napier to Palmerston North | 190km         | 2 hours 45 minutes |                      |
| <b>whole trip:</b>         |               |                    |                      |

Complete the table above and distance-time graph below.



- (d) The paving around a garden changes width on each side, given in the table below.

The central garden is 5 metres by 18 metres (total 90 square metres).

|        |     |      |        |       |
|--------|-----|------|--------|-------|
| side:  | top | left | bottom | right |
| width: | $x$ | $2x$ | $3x$   | $4x$  |



- (i) Ash writes the area of paving as  $A = (5 + 4x)(18 + 6x) - 90$ .

Benji write the area of paving as  $B = 6x(4x + 17)$ .

Show that these two representations are equal.

- (ii) Carol wants to find width  $x$  which makes the shaded area of paving also have an area of  $90 \text{ m}^2$ .

She does the following algebraic working:

$$\begin{aligned} 6x(4x + 17) &= 90 \\ 24x^2 + 102x - 90 &= 0 \\ (3x + 15)(8x - 6) &= 0 \end{aligned}$$

Show that her last line of working (the factorisation) is correct, and give the width  $x$  for her final answer.

## QUESTION TWO

- (a) A mathematical problem involves the following two equations.

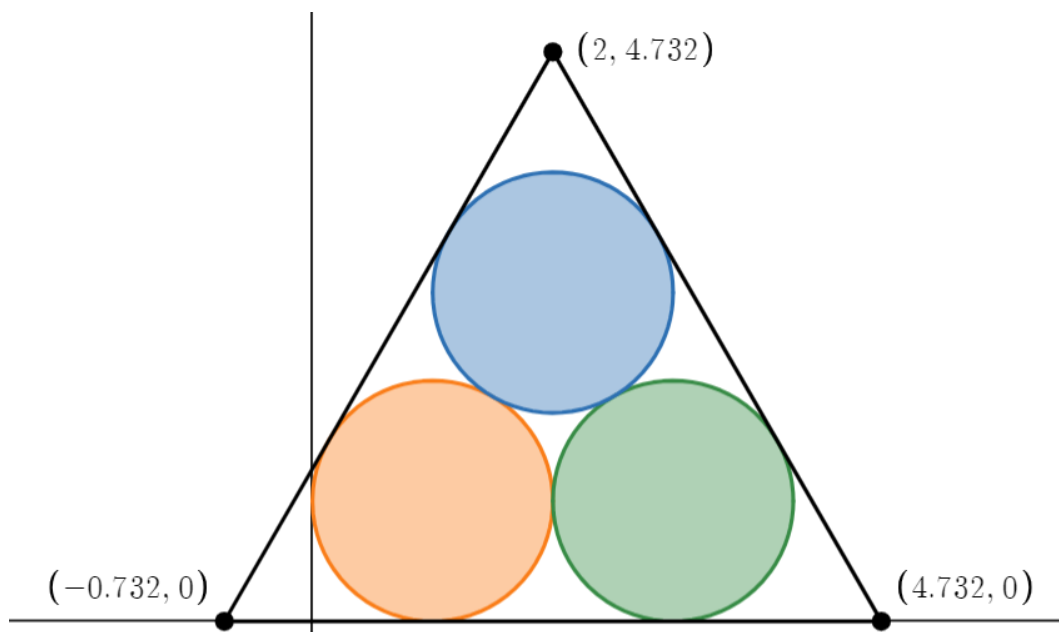
$$y = abd$$

$$Y = y \left( \frac{A}{a} + \frac{B}{b} + \frac{D}{d} \right)$$

Show that these can be rearranged to  $Y = Abd + aBd + abD$ .

*Be careful with uppercase and lowercase letters.*

- (b) Explain why a triangle with side lengths 28 mm, 96 mm and 100 mm is a right angled triangle, then calculate the angles at its corners.
- (c) The diagram below shows an equilateral triangle, with three circles of radius 1 cm drawn inside. The  $(x, y)$ -coordinates of the triangle corners are given.



Calculate the area of the triangle not covered by the shaded circles.

- (d) Josh is designing a cylindrical container that is closed at both ends with radius 8 cm. He needs the area of one of the end circles to be 5% of the total surface area of the container.



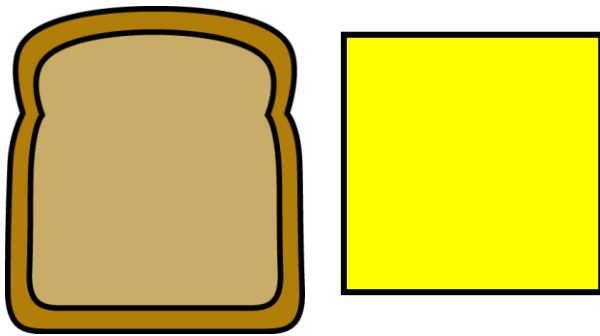
Check whether the container can hold 4 litres.

- (e) The image below shows a slice of bread and a slice of cheese.

The total cross-sectional area of the bread is  $135 \text{ cm}^2$ , and it weighs 38 grams. It is 9 mm thick. The outer crust makes up 30% of the bread by volume, and 45% of the bread by weight.

Also shown is 24 gram slice of cheese. The cheese is a 9 cm square, and is 3 mm thick.

The image is only approximately to scale.



Ivan's mother makes him a sandwich with two slices of bread that have the crusts removed and one slice of cheese.

Show that calculating the proportion of the sandwich which is bread by **weight** and by **volume** give quite different answers.

### QUESTION THREE

- (a) The logo for the band **Def Leppard** is shown below.



The outside of triangular letter D is an isosceles triangle 8 mm wide and 34 mm tall.

- (i) Find the obtuse angle in the letter D.

- (ii) The triangle inside the letter D has the same internal angles. It takes up 20% of the area of the triangle. Find the dimensions of the smaller triangle.
- (b) The diagram below shows a sequence of triangles. All have the same height, while the base is half the previous length at each step.

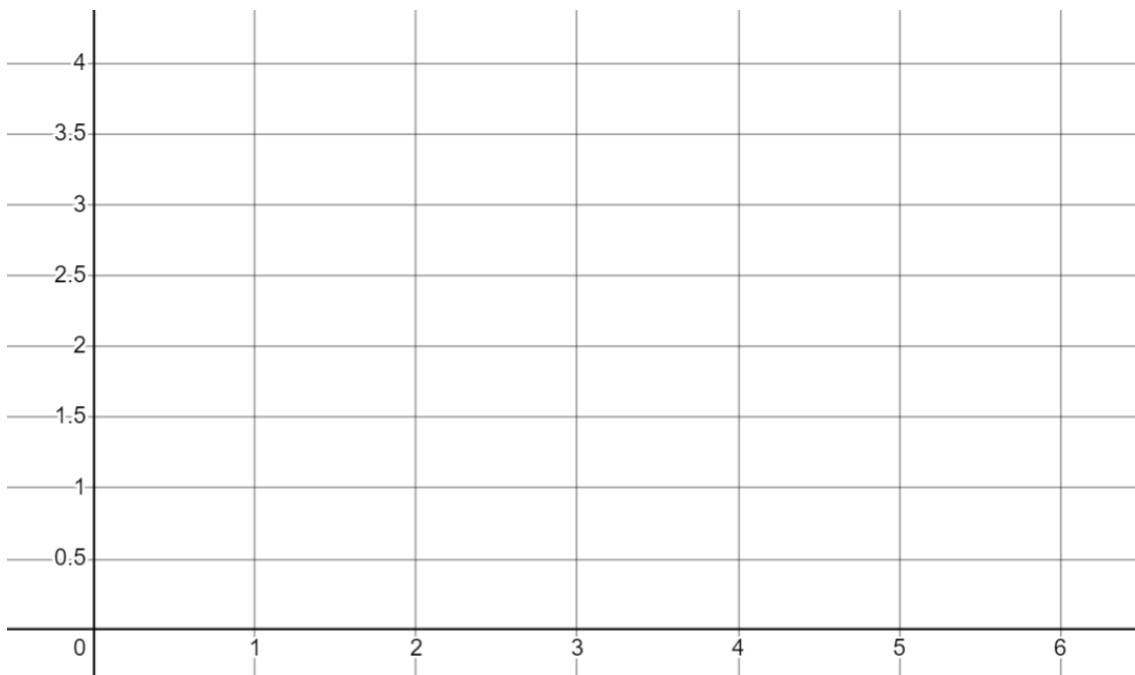
The third triangle has base and height both 1cm.



- (i) Find the angles in the first triangle.
- ii) Sketch a graph of the areas of the triangles, with  $n$  (the number of the triangle in the sequence) on the horizontal axis.

Give an equation for the area  $A$  in terms of  $n$ .

| triangle number $n$ | area $A$ |
|---------------------|----------|
|                     |          |
|                     |          |
|                     |          |
|                     |          |
|                     |          |
|                     |          |



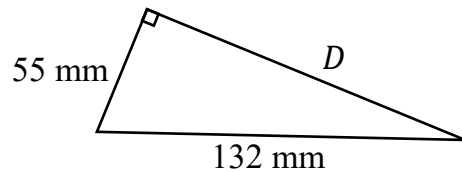
- (c) There are five consecutive positive integers that when squared and added satisfy the following equation. Here  $x$  represents the smallest integer.

$$x^2 + (x + 1)^2 + (x + 2)^2 = (x + 3)^2 + (x + 4)^2$$

Use algebra to find the value of  $x$ .

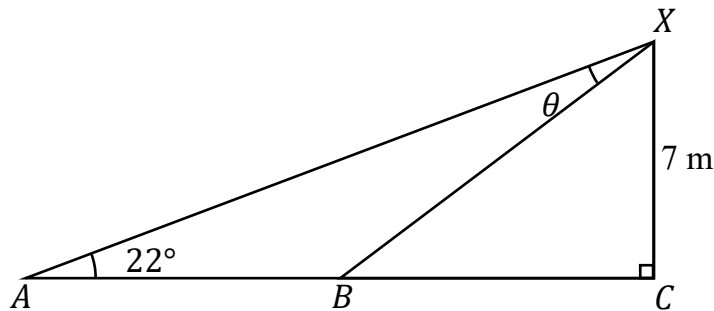
**MATHEMATICAL REASONING 91947-EXM-2**
**SNEDMATHS**
**QUESTION ONE**

- (a) Find the length  $D$  in the diagram below.



- (b) The point  $B$  in the diagram below is halfway between  $A$  and  $C$ .

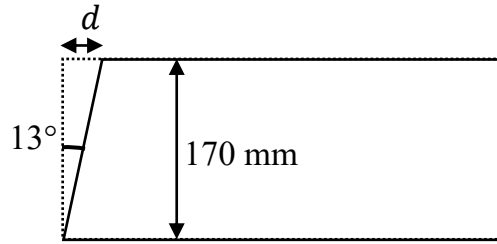
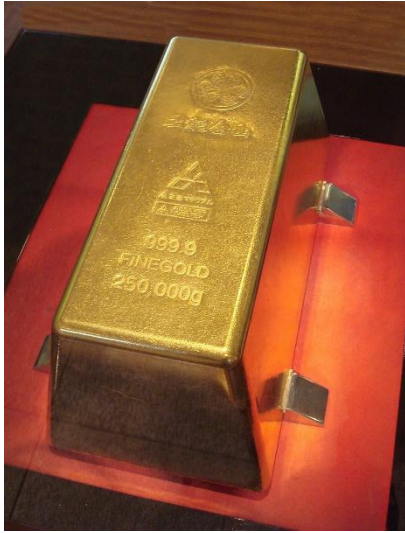
Find the angle  $\theta = \angle AXB$ .



- (c) Solve the inequality  $x^2 \leq 4x - 3$ .
- (d) The world's largest bar of gold is shown below. The base of the bar is 455 mm by 225 mm, and the block is 170mm high. It weighs 250 kg.

The four sides of the block slope inwards slightly with a *draft angle* of 13 degrees.

The draft angle is shown in the side-view on the right.



- (i) Calculate the distance  $d$  in the diagram above.
- (ii) The surface area of the top surface is  $S = (455 - 2d)(225 - 2d)$ .  
Expand the expression for  $S$ .
- (iii) The volume of the brick can be estimated using the formula

$$V = \frac{1}{3}h(S + \sqrt{SB} + B)$$

where  $S$  is the area of the top surface,  $B$  is the area of the base surface, and  $h = 170$  (millimetres) is the height of the brick.

Calculate the volume of the brick, and hence estimate the mass of one cubic centimetre of gold.

## QUESTION TWO

- (a) Expand and simplify

$$(x + y)(x^2 - xy + y^2)$$

- (b) Simplify and solve

$$\frac{x^2 - 9}{x^2 - 5x + 6} = 2$$

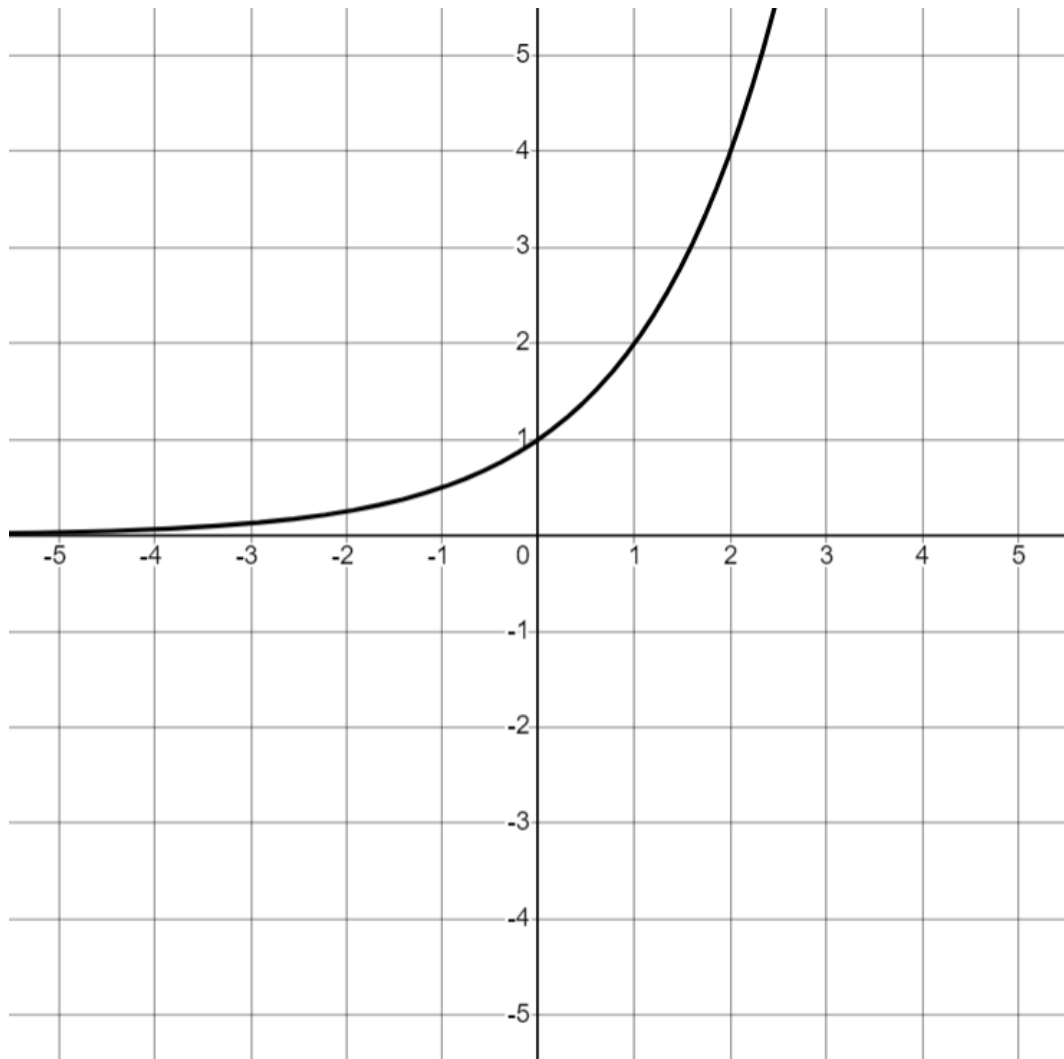
- (c) A florist sets the price of delivering a bunch of  $N$  roses as

$$C = 7N + 13$$

- (i) How much will it cost to have 12 roses delivered?
- (ii) How many roses can be delivered for \$200?

- (d) The graph below shows the relationship  $y = 2^x$ .

Sketch on the same axes the relationship  $y = 2^{x-3}$ , and explain how subtracting 3 in the exponent moves the curve.



- (e) A common student mistake is to forget to use brackets in algebraic expressions.

For example, the following two equations are not equivalent.

$$y = (1 - 2x)^2$$

$$y = 1 - 2x^2$$

Find the points where these different parabolas intersect.

**Explain your method clearly.**

### QUESTION THREE

- (a) A wall in the shape of a trapezium has height  $h$ , and top and bottom lengths  $a = 12 - x$  and  $b = 12 + x$ , where  $x$  is an unknown distance.

Show that the area of the wall does not depend on  $x$ .

- (b) Two streetlights are distance  $x = 15$  metres and  $y = 20$  metres from a door.

Adding the brightness of these streetlights gives a total brightness equivalent to one streetlight at a distance of  $d$  from the door using the equation below.

$$\frac{b}{d^2} = \frac{b}{x^2} + \frac{b}{y^2}$$

Find the distance  $d$ .

- (c) The following calculation is correct (to three significant figures).

$$4^{3.822} = 200$$

Find the value of  $x$  so that  $4^x = 800$ .

- (d) A movie theatre sells adult and child tickets.

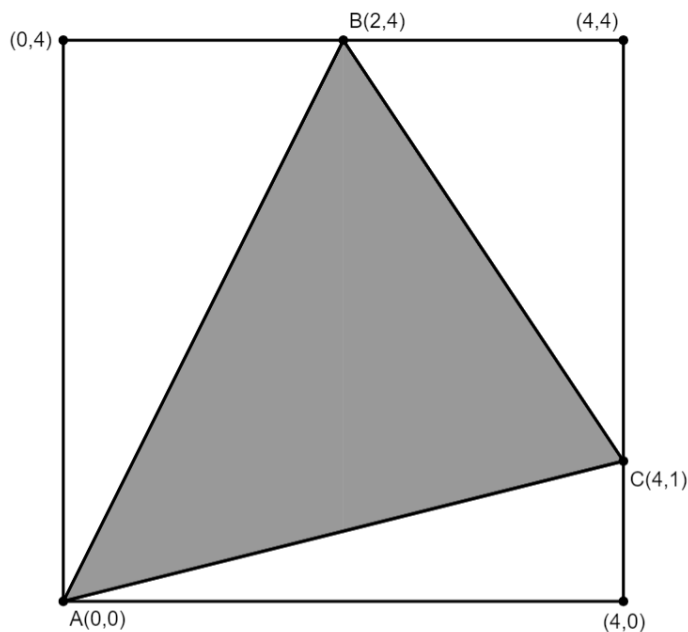
A group of 5 adults and 1 child spends \$111 on tickets.

A group of 3 adults and 3 children spends \$99 on tickets.

Use algebra to find the number price of each type of ticket.

- (e) The shaded triangle below is inside a square with side length 4cm.

Find the angles at the corners of the triangle ABC.



- (f) A parabola passes through the points  $(5,0)$ ,  $(7,0)$  and  $(9,11)$ .

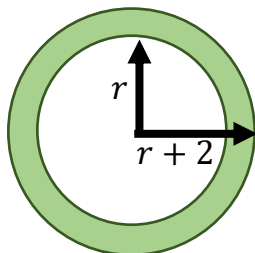
Use algebraic methods to find where the parabola intersects with the  $y$ -axis.

**MATHEMATICAL REASONING 91947-EXM-3**

**SNEDMATHS**

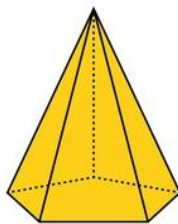
**QUESTION ONE**

- (a) A circular track has an inner radius of  $r$  metres and a constant width of 2 metres all the way around.



Write an expression for the area of the track in terms of  $r$  and simplify it.

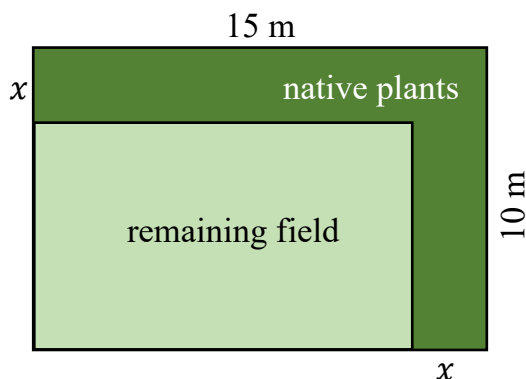
- (b) A prism with base area  $250 \text{ cm}^2$  is shown below.



The volume of the pyramid is one litre ( $1000 \text{ cm}^3$ ). Find the height of the pyramid.

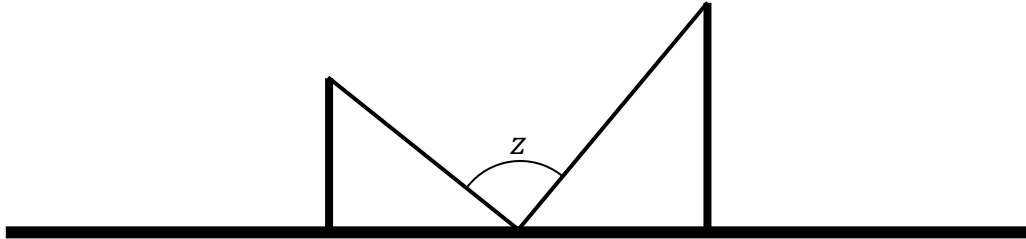
- (c) A rectangular field has two strips of native plants along the edges as shown.

Show that the areas of the remaining field is  $A = x^2 - 25x + 150$ , and hence find the value of  $x$  when the area is 84 square metres.



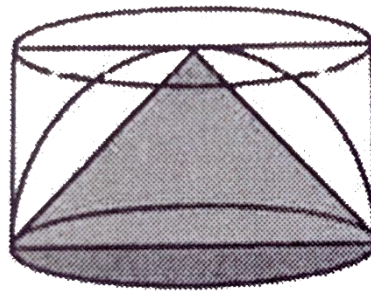
- (d) Two vertical poles stand on level ground 20 m apart. The poles are 8m high and 12m high.

A stake is placed somewhere on the ground half-way between the poles, and a rope is tied between the stake and the tops of the two poles as shown.



Find the angle between the ropes at the stake,  $z$ .

- (e) A cone fits perfectly inside a hemisphere which fits perfectly inside a cylinder, all of the same base radius  $R$ . They all have the same height,  $H = R$ .



- (i) Form and simplify the ratio of the **volumes** so that the terms are integers.

volume of cone : volume of hemisphere : volume of cylinder

$$a : b : c$$

- (ii) Another equation for cones gives the surface area,  $S = \pi r(r + \sqrt{r^2 + h^2})$ .

Rearrange this equation to find an expression for  $h$  in terms of  $r$  and  $S$ .

Expand and simplify your answer if possible.

## QUESTION TWO

- (a) A 3 metre long ladder leans against a wall. The base of the ladder is 1.1 metres away from the wall. Calculate how high up the wall the ladder reaches.



- (b) Solve the inequality  $5x^2 - 16x + 3 \leq 0$ .

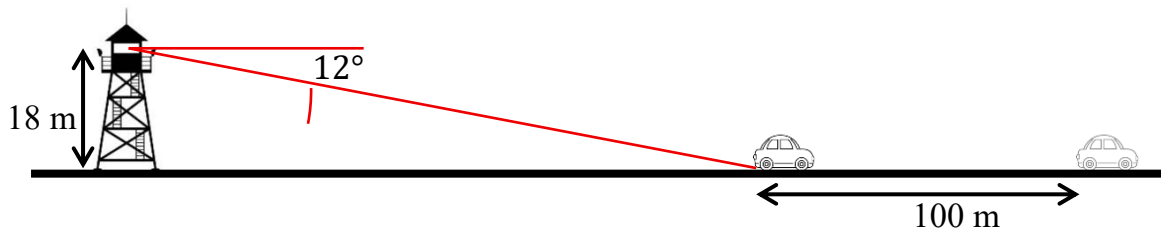
- (c) A straight line passes through the  $(x, y)$  points  $(1, 4)$  and  $(5, -2)$ .

Find the equation of the line in the form  $y = mx + c$ .

- (d) From a lookout tower with height 18 metres, the angle of depression to a car on a level road is  $12^\circ$ .

Find the new angle of depression when the car moves a further 100m away from the tower.

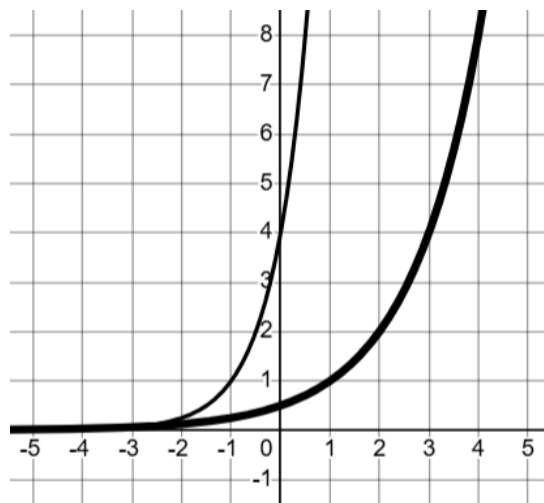
The diagram below represents the situation but is not to scale.



- (e) The two curves given are shown on the axes below.

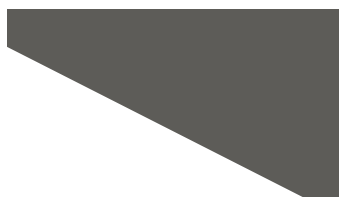
$$y = 2^{x-1}$$

$$y = 4^{x+1}$$



Use algebraic methods to find the coordinates where the curves intersect.

- (f) A rectangular sheet of metal  $x + 2$  by  $x + 3$  metres has a triangular corner removed, with side lengths  $x + 1$  and  $x + 2$  metres as shown.



Find the dimensions of the sheet of metal if the total area is 20 square metres.

### QUESTION THREE

- (a) Simplify

$$\frac{x^2 + 6x + 5}{x^2 + 8x + 7}$$

- (b) Rearrange  $v^2 = u^2 + 2as$  to make  $u$  the subject.

- (c) Two numbers add to 52.

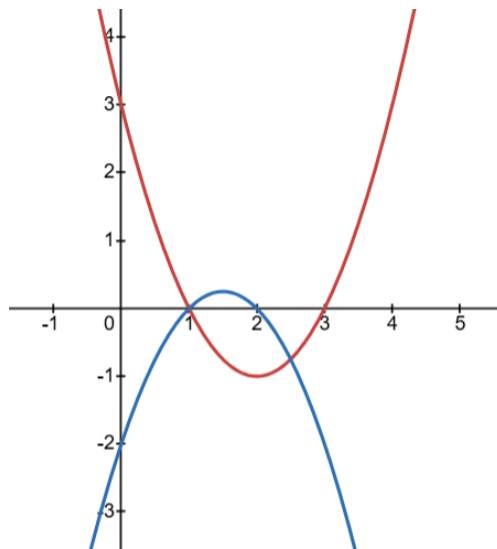
The larger number is 7 more than twice the smaller number.

Find both numbers.

- (d) The first four terms in a **quadratic** pattern are 3, 7, 13, 21.

Use algebra to find a rule for the  $n$ th term.

- (e) Two parabolas are shown below.



- (i) For each, find its equation and vertex.

Hence, find the distance between the vertices.

- (ii) The shape in the middle is used to make part of a company logo. A line between the intersection points is added.

Find an equation for the line.



## QUESTION ONE

- (a) A cylinder is twice as tall as it is wide.

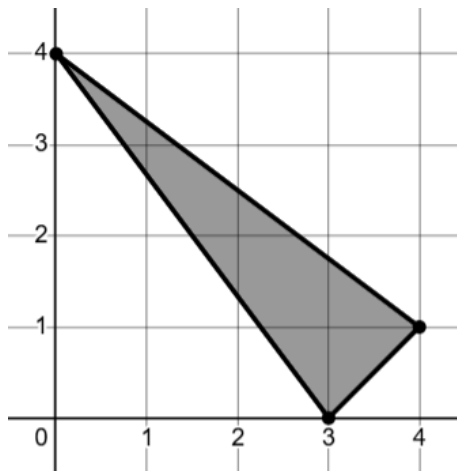
Write an expression for the volume in terms of the radius  $r$ , and simplify it where possible.

- (b) A rectangular poster has dimensions 40 by 60 cm. A uniform border of width  $x$  is left blank around the image on the poster.



Write an expression for the area of the image in terms of  $x$ .

- (c) Find the perimeter of the triangle shown in the diagram.

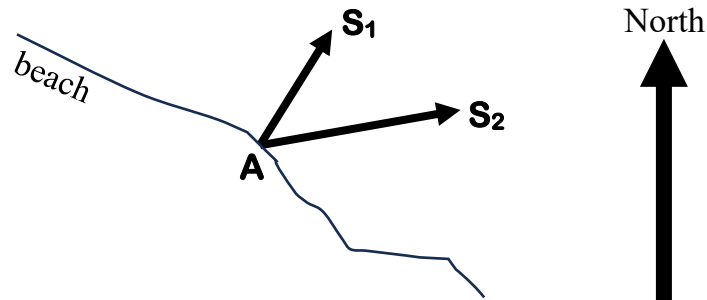


- (d) Astoria stands on a beach at point **A** and can see two ships.

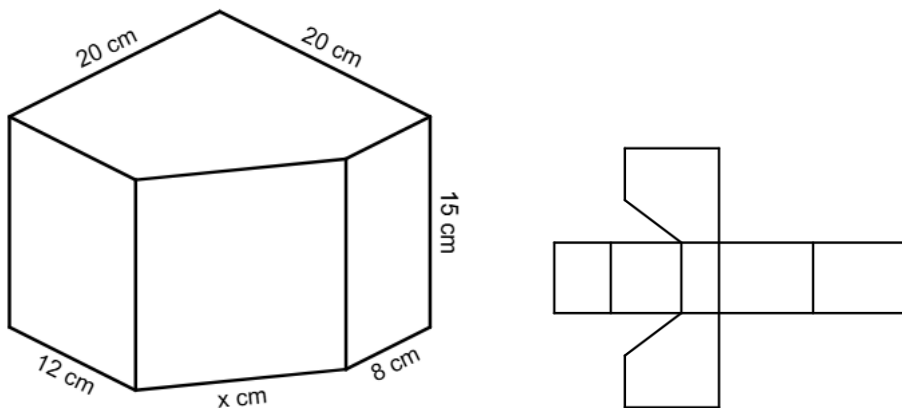
The first ship (**S<sub>1</sub>**) is 2km away at a bearing of 032°.

The second ship (**S<sub>2</sub>**) is 3km away at a bearing of 080°.

Find the distance between the two ships, and the bearing from **S<sub>1</sub>** to **S<sub>2</sub>**.



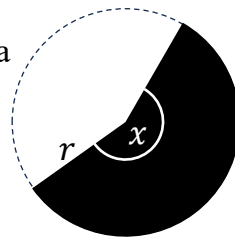
- (e) A rectangular block starts 20 by 20 by 15 centimetres, and has a side cut off to make the prism below. Its net is shown to the right.



Find the percentage that the surface area of the block has been reduced.

- (f) The area of a sector of a circle is calculated with the formula

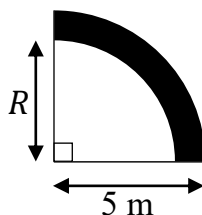
$$A = \frac{\pi x r^2}{360}$$



where  $r$  is the radius, and  $x$  is the angle (in degrees) at the centre of the circle.

The area of the shaded shape shown below is  $2\pi \text{ m}^2$ .

Find the inner radius  $R$ .



## QUESTION TWO

- (a) Solve the inequality  $2x + 7 \leq 4x + 17$ .

- (b) A straight line passes through (1,5) and (4,1).

Use algebraic methods to write an equation for the line through these points.

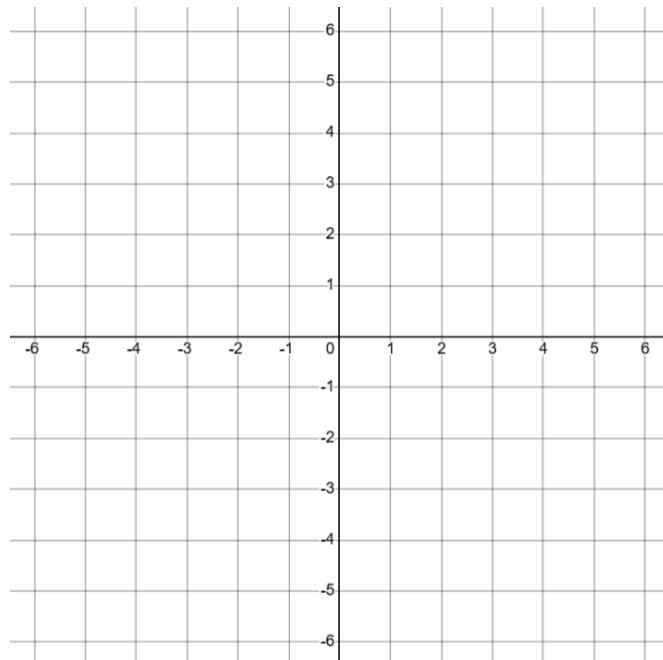
Write your answer in the form  $ax + by = c$  where  $a$ ,  $b$  and  $c$  are integers.

- (c) A simple model for the temperature  $T$  °C of a liquid is  $T = 80 - 5t$ , where  $t$  is the time in minutes after it is removed from a heating element.

Use algebraic reasoning to find when the temperature reaches 50 °C.

- (d) Sketch the parabola  $y = x^2 - 6x + 5$ , labelling

- the vertex
- the axis of symmetry
- the  $x$ -axis intercepts
- the  $y$ -axis intercept



- (e) Solve  $(2x + 5)(x + 3) + 2x = 0$ .

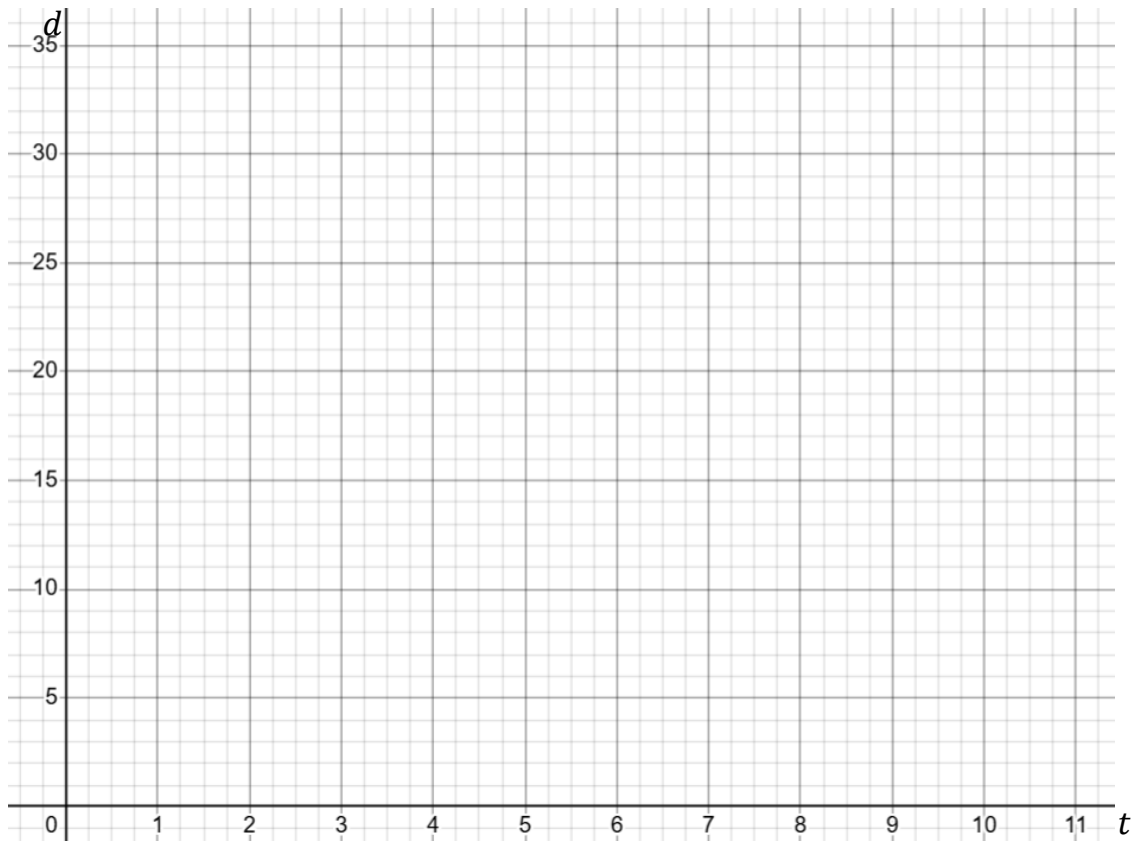
Use these  $x$ -values to find the **smallest** value of  $y = (2x + 5)(x + 3) + 2x$ .

- (f) A small insect travels north for 1 second at 1 cm per second, then for 2 seconds at 2 cm per second, then for 3 seconds at 3 cm per second, and finally for 4 seconds at 4 cm per second.

At time  $t = 0$ , the insect's position is  $d = 0$ .

- Sketch the position of the insect,  $d$ , at time  $t$  on the axes below.
- Find the time when the insect has travelled half the total distance.

Show algebraic working in your answer.



### QUESTION THREE

- (a) The first four terms in a sequence are 2, 6, 18, 54, ...

Find an equation for the  $n$ th term in the sequence.

- (b) The sides of a right-angled triangle are  $x$ ,  $x + 7$  and  $x + 8$  cm.

Form a quadratic equation and solve to find the value of  $x$ .

- (c) Andrea is 6 years older than her husband Blake. She also says she is 15% older than Blake.

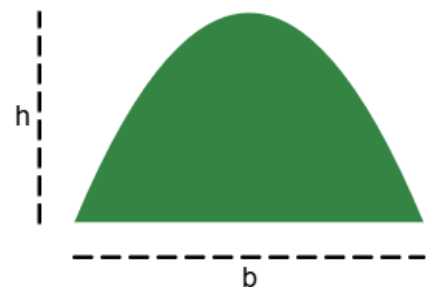
Write two equations that represent this information and solve for their ages.

- (d) The area of a parabolic segment can be calculated with the equation  $A = \frac{2}{3}bh$  where  $b$  is the base width and  $h$  is the height.

A parabolic segment has area  $165 \text{ cm}^2$ , and the base is 10% wider than the height.

Find the dimensions  $b$  and  $h$ .

Show algebraic working.

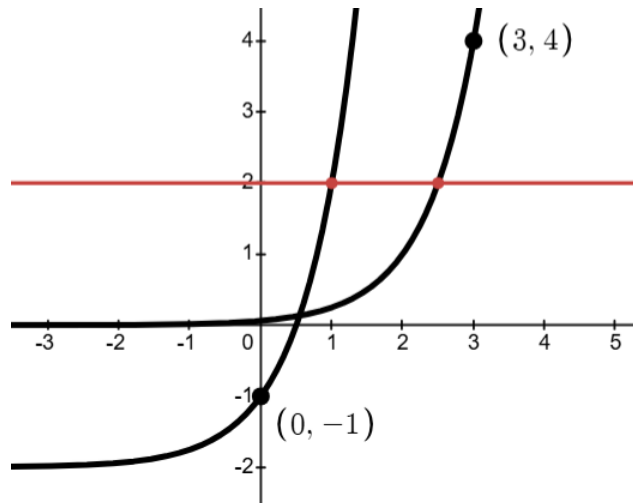


- (e) The graph below shows the exponential curves

$$y = 4^x - A$$

$$y = 4^{x-B}$$

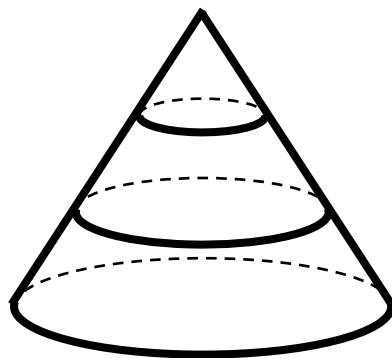
and the horizontal line  $y = 2$ .



Find the values of  $A$  and  $B$ , and the points where the horizontal line intersects with the two exponential curves.

- (f) A cone of height  $3H$  is cut horizontally to form three pieces as shown. Each piece has height  $H$ .

Use a radius of  $3x$  for the base of the whole cone.



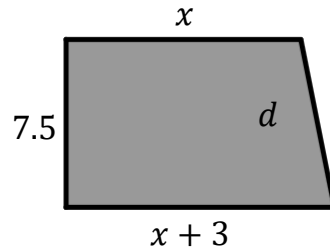
Find the ratio of the volume of the smallest piece to the volume of the largest, in the simplest form.

**QUESTION ONE**

- (a) Solve the inequality

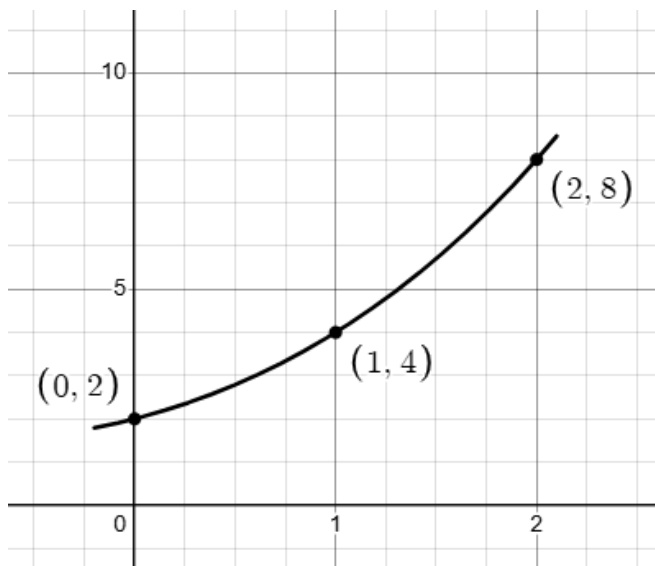
$$0.6x + 3 > 0.5x + 11$$

- (b) A straight line passes through the points  $(0,3)$  and  $(5,1)$ . Find the equation of the line in the form  $y = mx + c$ .
- (c) The floor plan of a classroom is in the shape of a trapezium as shown (the diagram not to scale). The dimensions are in metres.



Find the **total perimeter** for which the classroom floor area is 69 square metres.

- (d) The curve shown below could be modelled as either a parabola or an exponential.



$$y = 2^{x+p}$$

$$y = x^2 + x + q$$

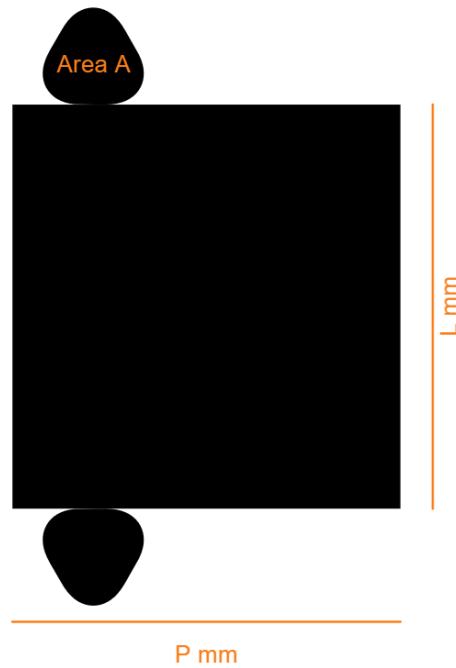
Find the values of  $p$  and  $q$  in these equations, and check these curves fit the points.

- (e) While using his calculator, Jonah notices that when two angles add to  $90^\circ$ , the cosine of one angle is the sine of the other.

For example,  $\cos 43^\circ = 0.682 = \sin 47^\circ$ .

Use a right-angle triangle to explain why this is always true.

- (f) The cross-sectional area of a piece of licorice candy is  $A = 20 \text{ mm}^2$ , and the perimeter of this area is  $P = 24 \text{ mm}$ . It is as a prism with a rounded triangle cross-section. The net is shown below (not to scale).



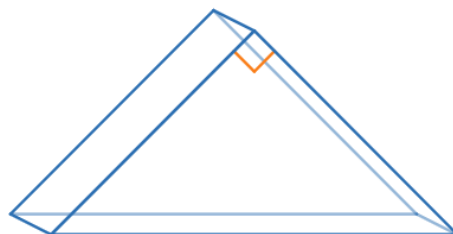
The total surface area is  $640 \text{ mm}^2$ . Calculate the volume.

## QUESTION TWO

- (a) Use algebraic reasoning to find the point where the lines  $y = 3x - 11$  and  $y = \frac{1}{2}x + 4$  intersect.
- (b) A glass triangular prism is shown below. The triangular faces are right-angled, with two sides length  $45 \text{ mm}$ . The prism is  $9 \text{ mm}$  thick.

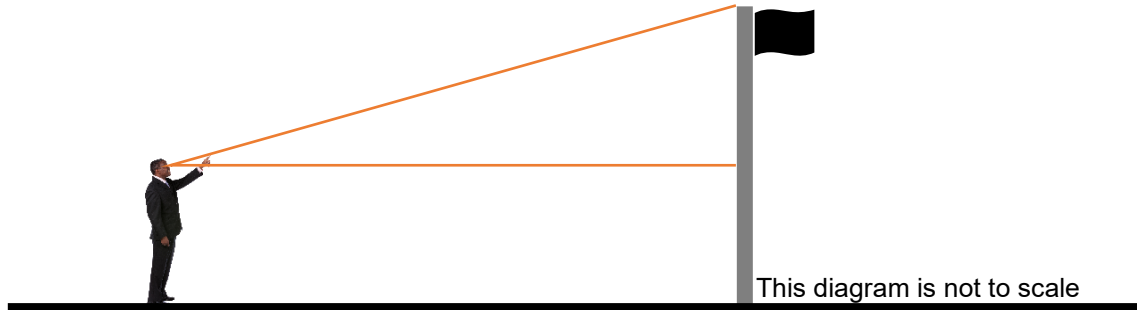
One cubic centimetre of glass has a mass of  $2.525 \text{ grams}$ .

Calculate the mass of the glass prism.

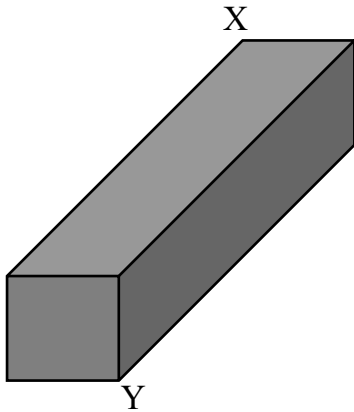


- (c) Caleb stands 25 metres away from a flagpole, and uses a clinometer to measure the angle of elevation to the top of the flagpole as  $18.5^\circ$ . Caleb's eyes are 1.64 metres above the level ground.

What percentage of the flagpole is below Caleb's eye level?



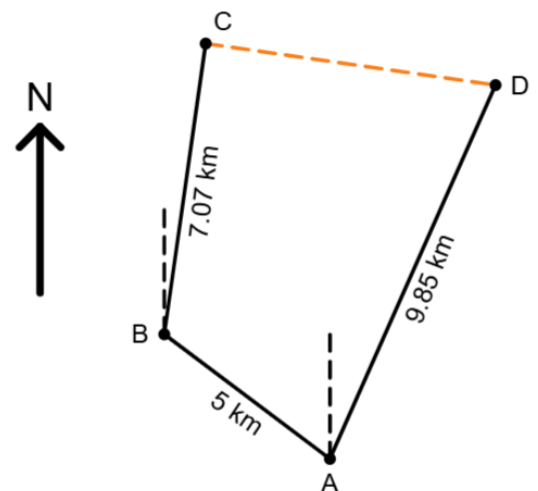
- (d) A box with square ends 1.7 metres long, and the long diagonal inside the box (from corner X to corner Y) is 1.9 metres.



Calculate the volume of the box.

- (e) The diagram shows four points in a national park. Three bearings are known between these points.

- from A to D, 9.85 km bearing  $024^\circ$
- from A to B, 5 km bearing  $307^\circ$
- from B to C, 7.07 km bearing  $008^\circ$



Find the distance and bearing from C to D.

- (f) Solve the equation below for  $x$ .

$$b^{-8}(ab)^x = \left(\frac{a}{b}\right)^x.$$

Show clearly your algebraic working.

### QUESTION THREE

- (a) The Great Pyramid of Giza is 138.5 metres tall, and the perimeter of the square base is 921 metres.

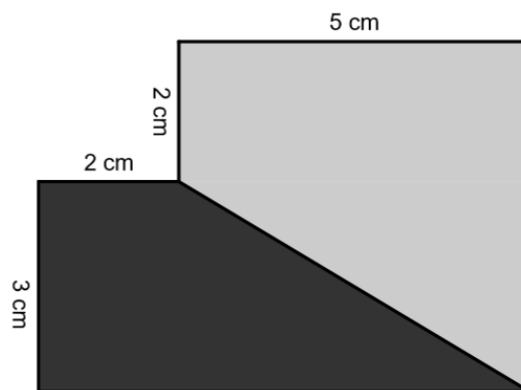
Calculate its volume in cubic metres.



- (b) The first four terms of a sequence are 7, 11, 15, 19, ...

Find the 16<sup>th</sup> term of the sequence.

- (c) Find the ratio of the shaded areas (dark : light) of the diagram below.

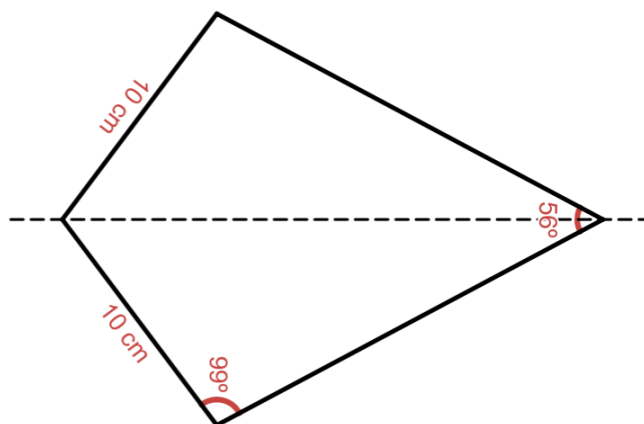


Write the ratio in integer form.

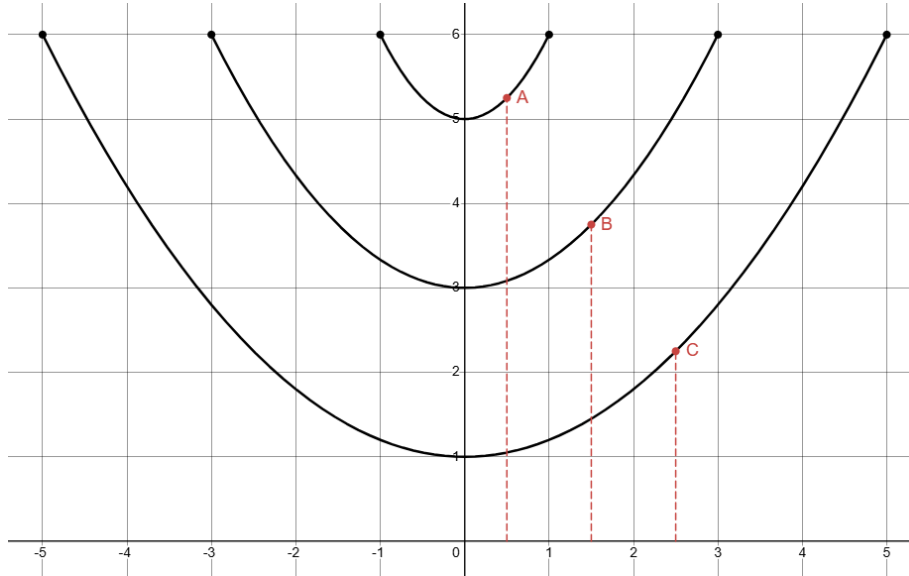
- (d) Solve the equation  $x(2x + 1) = 10$ .

Show algebraic working in your answer.

- (e) Find the perimeter and area of this kite. The dashed line is an axis of symmetry.



- (f) The graph below shows three parabolas and three points on these parabolas.



The parabolas have equations:

$$y = x^2 + 5$$

$$y = kx^2 + 3$$

$$y = \frac{1}{5}x^2 + 1$$

The points shown on the parabolas are:

$$A = (0.5, 5.25)$$

$$B = (1.5, b)$$

$$C = (2.5, c)$$

- Find the values of  $k$ ,  $b$  and  $c$ .
- Determine whether  $B$  is halfway between  $A$  and  $C$ .

Thanks to E Sneddon, R Fowler and M Horwood for proofreading and suggestions.

The cover image is of a Haast eagle and a moa. Both are extinct; just as NCEA will soon be extinct.